

Disproof of conjecture 00000000427

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Abstract

Conjecture 00000000427 asserts that the count of spin symmetric plane partitions at $t = 1$ is given by $F(n) = 2^{\lfloor n^2/4 \rfloor} \prod_{i=1}^n (2i-1)!!/i!$. We show that $F(n)$ is a non-integer rational number for every $n \geq 4$; in particular $F(4) = 525/2$. Since a count at $t = 1$ is a cardinality, and hence a non-negative integer, the conjecture is false already at $n = 4$. An independent exhaustive enumeration of order ideals of $[n]^3$ is included as supporting evidence, and does not enter the proof.

1 The conjecture

Definition. Spin symmetric plane partitions are plane partitions in a cubic box symmetric under all diagonals and weighted by spin.

Conjecture. Their count at $t = 1$ equals

$$2^{\lfloor n^2/4 \rfloor} \prod_{i=1}^n \frac{(2i-1)!!}{i!},$$

and the ratio of this formula to the unsymmetrized MacMahon formula is exactly a product of a power of two and double factorials, with no other prime factors.

The conjectured formula, read as an exact rational number, is the object we analyse.

2 The formula is not an integer

Definition 1. The *odd double factorial* is $(2i-1)!! = \prod_{j=1}^i (2j-1)$ for $i \geq 1$, with $(2 \cdot 0 - 1)!! = 1$. For $n \geq 0$ define

$$F(n) = 2^{\lfloor n^2/4 \rfloor} \prod_{i=1}^n \frac{(2i-1)!!}{i!} \in \mathbb{Q}.$$

Write ν_2 for the 2-adic valuation of a non-zero rational, and set $D(n) = \sum_{i=1}^n \nu_2(i!)$.

Lemma 2. For every $n \geq 0$,

$$\nu_2(F(n)) = \left\lfloor \frac{n^2}{4} \right\rfloor - D(n).$$

Proof. Every $(2i-1)!!$ is a product of odd integers, hence odd, so $\nu_2((2i-1)!!) = 0$. The valuation is additive on products and satisfies $\nu_2(x/y) = \nu_2(x) - \nu_2(y)$ for non-zero rationals x, y . Therefore

$$\nu_2\left(\prod_{i=1}^n \frac{(2i-1)!!}{i!}\right) = \sum_{i=1}^n (0 - \nu_2(i!)) = -D(n),$$

and, since the leading factor $2^{\lfloor n^2/4 \rfloor}$ is a power of two,

$$\nu_2(F(n)) = \left\lfloor \frac{n^2}{4} \right\rfloor + (-D(n)) = \left\lfloor \frac{n^2}{4} \right\rfloor - D(n).$$

This is the claim. □

Theorem 3 (The counterexample $n = 4$). $F(4) = 525/2$, which is not an integer. In particular $\nu_2(F(4)) = -1$.

Proof. We compute

$$D(4) = \nu_2(1!) + \nu_2(2!) + \nu_2(3!) + \nu_2(4!) = 0 + 1 + 1 + 3 = 5,$$

while $\lfloor 4^2/4 \rfloor = 4$. By Lemma 2,

$$\nu_2(F(4)) = 4 - 5 = -1 < 0,$$

so $F(4)$ has a factor of 2 in its denominator in lowest terms, hence is not an integer. Directly,

$$F(4) = 2^4 \cdot \frac{1 \cdot 3 \cdot 15 \cdot 105}{1! \cdot 2! \cdot 3! \cdot 4!} = 16 \cdot \frac{4725}{288} = \frac{75600}{288} = \frac{525}{2}. \quad \square$$

Theorem 4. *For every $n \geq 4$, $F(n)$ is not an integer; more precisely $\nu_2(F(n)) < 0$.*

Proof. Put $G(n) = \lfloor n^2/4 \rfloor - D(n) = \nu_2(F(n))$ by Lemma 2. We first compute the increment. Since $n^2 - (n-1)^2 = 2n - 1$ one checks

$$\left\lfloor \frac{n^2}{4} \right\rfloor - \left\lfloor \frac{(n-1)^2}{4} \right\rfloor = \left\lfloor \frac{n}{2} \right\rfloor,$$

and $D(n) - D(n-1) = \nu_2(n!)$. Legendre's formula gives $\nu_2(n!) = \sum_{j \geq 1} \lfloor n/2^j \rfloor$. Hence

$$G(n) - G(n-1) = \left\lfloor \frac{n}{2} \right\rfloor - \nu_2(n!) = - \sum_{j \geq 2} \left\lfloor \frac{n}{2^j} \right\rfloor \leq 0.$$

For $n \geq 4$ we have $\lfloor n/4 \rfloor \geq 1$, so $G(n) < G(n-1)$: the sequence G is strictly decreasing from $n = 4$ onward. Finally $G(3) = \lfloor 9/4 \rfloor - D(3) = 2 - (0 + 1 + 1) = 0$, so $G(4) = -1$ and by induction $G(n) \leq -1$ for all $n \geq 4$. Thus $\nu_2(F(n)) = G(n) < 0$ for every $n \geq 4$. $\square$

Corollary 5. *No natural number equals $F(4) = 525/2$. More generally, the conjectured equality between $F(n)$ and the count of objects at $t = 1$ fails for every $n \geq 4$, and in particular at $n = 4$. Conjecture 00000000427 is therefore false.*

Proof. A count at $t = 1$ is the cardinality of a finite set, hence a non-negative integer. But $F(4) = 525/2 \notin \mathbb{N}$ by Theorem 3, and $F(n) \notin \mathbb{Z}$ for all $n \geq 4$ by Theorem 4. Hence no such count can equal the conjectured formula. $\square$

Remark 6. For $n = 1, 2, 3$ the formula is integral, $F(1) = 1$, $F(2) = 3$, $F(3) = 15$, which is presumably why the claim looked plausible; integrality fails at $n = 4$ and, by Theorem 4, never recovers. The valuation mechanism is that $\nu_2(n!) = n - s_2(n)$ (where $s_2(n)$ is the number of ones in the binary expansion of n) grows faster than $\lfloor n/2 \rfloor$, so the exponent of 2 in the denominator of $F(n)$ strictly outruns the explicit power $2^{\lfloor n^2/4 \rfloor}$.

Remark 7. The secondary claim about the ratio to MacMahon's formula also fails under the natural reading of "a product of a power of two and double factorials" as a non-negative integer: at $n = 4$, using the MacMahon total $\prod_{i,j,k=1}^4 \frac{i+j+k-1}{i+j+k-2} = 232848$, the ratio is

$$\frac{F(4)}{232848} = \frac{525}{2 \cdot 232848} = \frac{525}{465696},$$

whose numerator is odd while its denominator is even. The main refutation, however, rests only on the count equality.

3 Independent enumeration (supporting evidence)

As a cross-check independent of the valuation argument, we exhaustively enumerated order ideals of the product poset $[n]^3$ (equivalently, plane partitions in the n -cube) and counted those invariant under all permutations of the three coordinates.

n	order ideals of $[n]^3$	invariant under all coordinate permutations
1	2	2
2	20	5
3	980	16
4	232848	66

The MacMahon totals 2, 20, 980, 232848 reproduce the classical product formula and validate the enumerator. The invariant counts 2, 5, 16, 66 disagree with the conjectured formula $1, 3, 15, 525/2$ already at $n = 1$ (1 vs 2) and $n = 2$ (3 vs 5), and at $n = 4$ the conjectured value is not even an integer.

The exact definition of “spin symmetric plane partition” is not pinned down by the conjecture, so this enumeration is offered as supporting evidence and not as the definition. The refutation does not rely on it: whatever the intended object, its $t = 1$ value is a count, and no count equals $525/2$.

4 Conclusion

The conjectured formula $F(n)$ is not an integer for any $n \geq 4$: already $F(4) = 525/2$, and $\nu_2(F(n)) < 0$ for all $n \geq 4$. A count at $t = 1$ is a non-negative integer, so the conjectured equality is false. Conjecture 00000000427 is disproved.